



NHF&RI Doctor's Duty Roster App

User Manual

National Heart Foundation Hospital
and Research Institute

Version 1.0 • Developed by Dr. Alif



Contents

- | | | |
|-----------|--|---|
| 1 | Introduction | What the app is and who it's for |
| 2 | Getting Started & Logging In | Master and roster-maker passwords |
| 3 | The Dashboard | Reading your setup at a glance |
| 4 | Managing Doctors | Categories, targets, wards, and the secret team |
| 5 | Wards | Adding and grouping wards |
| 6 | Shift Requirements (Stations) | How many doctors each station needs |
| 7 | Demands | Off, double, single, leave, and fixed assignments |
| 8 | Holidays & Events | Government holidays and automatic Fridays |
| 9 | Generating & Exporting the Roster | Build, review, save, and export as PDF |
| 10 | Summary | Per-doctor totals against target |
| 11 | Duty Bank & Fairness | Carry-over and Friday-night balancing |
| 12 | Notes | The shared scratchpad |
| 13 | Settings & Backup | Passwords, letterhead, and data export |
| 14 | How the Scheduling Algorithm Works | The rules behind fair rotation |
-



15 Tips, Warnings & Troubleshooting

Getting a clean roster

16 Glossary

Key terms in one place

SECTION 1

Introduction

The **NHF&RI Doctor's Duty Roster App** is a mobile-first web application for building and managing the monthly doctor duty roster at the National Heart Foundation Hospital and Research Institute. It replaces the slow, error-prone job of assembling a roster by hand with a smart scheduling engine that respects each doctor's category, ward restrictions, night and OPD limits, casual leave, and personal requests — while spreading duties fairly across the team.

What you can do with it

- Maintain a live list of doctors, their seniority categories, and duty targets.
- Describe how many doctors each ward/station needs, per shift.
- Record standing requests (“demands”): days off, double duties, casual leave, and fixed postings.
- Mark government holidays — every Friday is treated as a holiday automatically.
- Generate a complete monthly roster in one click, review any shortfalls, and export it as a PDF or print it.
- Track fairness over time with the **Duty Bank**, which carries over- and under-worked duties into next month.

Who uses it

There are two kinds of user. The **Master** (Dr. Alif) has full control, including managing the rotating passwords used by roster makers and the hidden “secret” duty team. A **Roster Maker** logs in with a time-limited password — typically issued for a three-month cycle — and can do everything needed to prepare and export a roster, but cannot see the secret team or manage passwords.

Works offline, syncs to the cloud

The app is a Progressive Web App (PWA). It caches itself so it keeps working with a weak connection, and it stores doctors, wards, demands, passwords, and saved rosters in a shared Supabase cloud database so every authorised user sees the same data.

SECTION 2

Getting Started & Logging In

Open the app in any modern browser on your phone, tablet, or computer. You can also “Add to Home Screen” to launch it like a native app. The first screen is always the login screen — there are no separate usernames, only a password that determines your role.

The two roles

Role	How you log in	What you can do
Master (Dr. Alif)	The fixed master password	Everything: manage doctors, the secret team, roster-maker passwords, letterhead, and generate/export rosters.
Roster Maker	A rotating password issued by the Master (may have an expiry date)	Manage doctors, wards, stations, demands, holidays; generate, save, and export rosters. Cannot see the secret team or manage passwords.

Logging in



- 1 On the login screen, type your password into the single password box. Tap the eye icon to reveal it if you want to check what you typed.
- 2 Tap **Unlock**. If the password matches the master password or an active maker password, you are taken straight to the Dashboard.
- 3 If a maker password has passed its expiry date, the app refuses it and asks you to contact Dr. Alif for a new one.

Signing out

Use **Sign Out** at the bottom of the sidebar (desktop) or from the **More** tab (mobile) whenever you finish on a shared device. Your role badge — “Master Access” or the maker label — is always shown so you know how you are logged in.

SECTION 3

The Dashboard

The Dashboard is the home screen after login. It shows a role badge and a grid of summary cards so you can confirm your setup at a glance before generating a roster.

Card	What it shows
Doctors	Number of active, non-secret doctors (plus a note if a secret team exists).
Wards	Number of active wards.
Demands	Total standing requests currently on file.
Holidays	Holidays marked for the selected month.
Roster	Whether a roster has been generated, and for which month.
Shortfalls	How many warnings the last generation produced — “All clear” when there are none.

On a computer, navigation lives in the left **sidebar**. On a phone, the five most common destinations — Home, Doctors, Roster, Summary, and More — sit in a bottom navigation bar.

SECTION 4

Managing Doctors

The **Doctors** page is where you keep the roster's roster of people. Search by name, filter by category, and tap **Add Doctor** to open the doctor form. Each doctor card shows their categories, monthly and night targets, ward count, and any duty date window.

The doctor form, field by field

Field	Meaning
Name	The doctor's display name, e.g. “Dr. Adan”.
Categories	One or more of SMO, EMO, MO, First Man. Category drives eligibility and default targets.
Monthly Target	How many duty slots this doctor should ideally work in the month. Defaults by category (SMO 18, EMO 22, MO 23).
Night Target	How many night shifts the doctor should get.

Field	Meaning
Cath-eligible	Whether the doctor can be posted to the Cath Lab, plus a monthly Cath quota.
Ward Restrictions	Optional. If set, the doctor is only placed at stations covering at least one of these wards.
OPD Min / Max	The floor and ceiling for OPD duties. Blank max means no cap. Defaults by category.
Duty Starts / Ends	Optional date window. Outside it, the doctor is treated as unavailable (e.g. a new joiner or someone leaving).
Secret team	Master only. Hides the doctor from roster makers (see below).

Categories at a glance

Category	Typical role	Default target	Default OPD
SMO	Senior Medical Officer	18	4–6
EMO	Emergency Medical Officer	22	0–4
MO	Medical Officer	23	0–(no cap)
First Man	Senior on-call, priority wards only	—	—

Activating, editing, and deleting

- **Edit** (pencil) reopens the form to change any field.
- **Off / On** toggles a doctor active or inactive. Inactive doctors are kept on file but are never scheduled — useful for someone temporarily unavailable.
- **Delete** (trash) permanently removes the doctor after a confirmation prompt.

The secret duty team (Master only)

Doctors flagged as “secret” are hidden from roster makers entirely. As Master you can reveal them on the Doctors page with **Show Secret Team**. This lets Dr. Alif maintain a confidential pool of doctors without exposing them to whoever is preparing the roster this cycle.

SECTION 5

Wards

Wards are the physical units that need cover. The **Wards** page lists them grouped into four families, and you can filter by group, add new wards, toggle them on/off, or delete them.

Group	Contains	Notes
General	Observation, 3A/3B, the 5-series, Wards 7/9/10/12, HTN, Cabin	The core inpatient wards.
DS	DS 15A/B/C, DS 9A/B, DS 8	Day-shift / short-stay units.
OPD	OPD A, OPD B, OPD C	Outpatient clinics — closed on holidays.
Cath	Cath (Cath Lab)	Only Cath-eligible doctors may be posted here.

Turning a ward **Off** keeps its history but removes it from new rosters and from the ward pickers used elsewhere. A ward's group also controls holiday behaviour: OPD, HTN, and Cath close automatically on

holidays.

SECTION 6

Shift Requirements (Stations)

A **station** is a staffing requirement: “this group of wards needs this many doctors on this shift.” The **Shift Requirements** page has a tab for each shift — Morning (08:00–14:30), Evening (14:30–21:00), and Night (21:00–08:00) — and lists the stations for that shift with a running total of slots.

Adding or editing a station

- 1 Pick the shift tab you want to staff.
- 2 Tap **Add Station** and give it a label (e.g. “Ward 9 & Cabin”).
- 3 Tick the wards this station covers — one station can span several wards.
- 4 Set how many doctors are needed, then Save.

The list is pre-seeded for you

The first time the app runs it loads a complete default set of stations for all three shifts (matching NHF&RI's real layout) into the database. You only need to adjust it when your staffing pattern changes — you are never starting from a blank page.

SECTION 7

Demands

Demands are standing requests the generator must respect. Add one from the Demands page by choosing a doctor, a kind, and when it applies. Filter the list by doctor to review someone's requests.

Kind	What it does
Off request	The doctor is not scheduled. Optionally limited to one shift; otherwise all shifts.
Double duty	The doctor works two back-to-back shifts — Morning+Evening or Evening+Night.
Single only	Overrides a standing double-duty for that day/week, forcing a single shift.
Casual leave	The doctor is unavailable across a date range. The generator avoids them, but may cover them as a last resort (and warns you).
Fixed assignment	Forces the doctor onto a specific ward and shift.

When a demand applies

- **Every week** — repeats on a chosen weekday (e.g. every Friday).
- **Specific date** — a day number of the month. Note: this repeats every month, so delete it once the month has passed.
- **Always** — available for double/single kinds, applies on every day.
- **Casual leave** always uses an explicit start and end date.

Housekeeping tip

“Specific date” demands are not tied to a particular month — a demand for “day 12” will fire on the 12th of every month until you remove it. Clear stale date-based demands at the start of each cycle.



SECTION 8

Holidays & Events

The **Holidays & Events** page records government holidays for a chosen month and year. On a holiday, OPD A/B/C, HTN, and the Cath Lab close for the day, and the generator trims the affected stations automatically.

- Select the month and year, tap **Add Holiday**, pick the day, and give it a label (e.g. "Independence Day").
- Holidays are listed for the selected month and can be deleted individually.
- **Every Friday is treated as a holiday automatically** — you never need to add Fridays yourself.

Why holidays matter

Closing OPD/HTN/Cath and thinning holiday staffing is how the roster reflects real hospital operations. It also frees doctors for the wards that stay open, which the fairness engine then balances.

SECTION 9

Generating & Exporting the Roster

The **Generate & Export** page is where everything comes together. A pre-flight panel shows the active doctor count, station count, number of demands, and holidays for the chosen month so you can sanity-check before building.

Step by step

- 1 Choose the **month** and **year**.
- 2 Tap **Generate Roster**. The engine assigns doctors across every day and shift in a moment. Re-tap **Regenerate** to get a fresh arrangement — there is deliberate randomness, so each run differs slightly.
- 3 Review the **warnings** panel (if any) — it lists shortfalls, skipped fixed assignments, leave that had to be covered, and doctors who missed a Friday night.
- 4 Use the **shift tabs** (Morning / Evening / Night) to read the day-by-day posting for each shift. Holidays are flagged inline.
- 5 Tap **Save Roster** to store this month's roster in the cloud and update the Duty Bank.
- 6 Tap **Export PDF** to download a print-ready PDF (one section per shift), or **Print** to use your browser's print dialog.

Before you can generate

You need at least one active doctor and one shift-requirement station. Until then the Generate button stays disabled and the page tells you what's missing.

What "Save" does beyond storing the roster

Saving also recomputes the **Duty Bank** for the month: for each doctor it records the base target, the effective target (reduced if they were over-worked the previous month), how many duties they were actually assigned, and the resulting balance. That balance is what feeds fairness into next month's generation.

SECTION 10

Summary

The **Summary** page reports per-doctor totals for the roster currently in memory. Search for a doctor, and read each row against their targets.

Column	Meaning
Assigned	Total duties assigned. A coloured (+/-) delta shows over/under the monthly target.
Target	The doctor's monthly target.
Nights	Night duties assigned vs. the doctor's night target.
Cath	Cath duties vs. quota, for Cath-eligible doctors.
OPD	OPD duties, shown against the OPD max where one is set.

Colour coding makes outliers obvious: **red** means under target (or below night target), and **orange** means over target. Use this page to decide whether to regenerate or hand-tune demands before saving.

SECTION 11

Duty Bank & Fairness

Fairness across months is the job of the **Duty Bank**. It is built automatically each time you save a roster, and you can browse any past month from the month selector.

Column	Meaning
Base Target	The doctor's normal monthly target.
Effective Target	The target after subtracting any overtime owed from the previous month.
Assigned	Duties actually worked that month.
Balance	Assigned minus effective target. Positive = over-worked (owed rest); negative = under-worked.

A doctor who runs **over** target this month has that many duties shaved off their effective target next month, so the load evens out over the cycle. The page also shows a **Friday Night Duties** breakdown: anyone with zero Friday nights this month is prioritised for one next month, and anyone who already did a Friday night is temporarily exempt from the next — so the least-popular slot rotates fairly.

SECTION 12

Notes

The **Notes** page is a shared scratchpad stored in the cloud and visible to everyone who logs in. Use it for anything worth remembering about the current cycle — for example, “Dr. Karim covering Dr. Rahim's Cath duties until the 20th.” Type your note and tap **Save Notes**; an “Unsaved changes” indicator reminds you when you have edits to save.

SECTION 13

Settings & Backup

The **Settings & Backup** page manages passwords, the printed letterhead, and data safety.

Roster-maker passwords (Master only)

- Tap **New Password**, enter the password text, a friendly **label** (e.g. “July–Sept 2026”), and an optional **expiry date**.
- Each password can be **copied**, **deactivated** (kept on file but no longer valid), or **deleted**.
- Issue a fresh password at the start of each cycle and let the old one expire — that is how access is rotated.

Letterhead

Set the **Hospital / Institute Name** and an optional **Prepared By** name. These appear at the top of every exported and printed roster. Changes are saved automatically.

Backup & Restore

Tap **Download Backup** to save a JSON snapshot of your doctors, wards, stations, demands, holidays, settings, and current month to your device — a useful safeguard before large changes.

Keep the master password safe

The master password grants complete control, including managing every roster-maker password. Only Dr. Alif should hold it, and it should be shared through a secure channel — never written on a shared roster or a public board.

SECTION 14

How the Scheduling Algorithm Works

You do not need to understand the internals to use the app, but knowing the rules helps you read warnings and shape good demands. For each day the engine walks Morning, then Evening, then Night, and for each shift it fills stations while enforcing these rules.

Hard rules (never broken)

- **One posting per shift** — a doctor cannot be in two places in the same shift.
- **Rest after nights** — a doctor who worked a night is not scheduled the very next day.
- **Category eligibility** — only Cath-eligible doctors go to the Cath Lab; First Man doctors are limited to their priority wards; ward restrictions are honoured.
- **Off requests and duty date windows** — respected absolutely.
- **Friday-night cap** — no doctor gets more than two Friday nights in a month.
- **Double-duty patterns** — back-to-back shifts only follow a valid Morning+Evening or Evening+Night request.

Fairness preferences (balanced, sometimes bent as a last resort)

- Spread duties toward each doctor's **effective target**, adjusted by the Duty Bank.
- Respect **night**, **Cath**, and **OPD** targets and quotas.
- Avoid running the same doctor on the same shift more than three times in a row, and keep a minimum gap between a doctor's duties based on their pace.
- Rotate the **Friday night** duty toward those who haven't had one.
- Avoid seating two senior/First-Man doctors in the same slot where a mix is preferred.

When there simply aren't enough eligible doctors, the engine relaxes the *soft* preferences one at a time (this is the “extra” pass) rather than leave a ward unmanned — and every time it does so, or still falls short, it records a warning for you to review.

Why two runs differ

The engine shuffles eligible doctors before ranking them, so regenerating produces a different but equally valid roster. If you get an arrangement you like, save it — regenerating replaces it.

SECTION 15

Tips, Warnings & Troubleshooting

Reading warnings

Warnings never block a roster — they flag where the ideal rules had to bend. Common ones:

Warning	What it means & what to do
"short by N doctor(s)"	A station couldn't be fully staffed. Add active doctors, relax a ward restriction, or reduce the station's requirement.
"fixed assignment ... skipped"	A fixed-assignment demand conflicted with a hard rule (rest, off, leave, eligibility). Adjust the demand.
"slot requiring [...] left empty"	No doctor of the required category was available for a composed slot (e.g. an SMO seat).
"covered by doctor on casual leave"	Staffing was so tight the engine had to use someone on leave. Consider adding cover or reducing requirements.
"did not get a Friday night duty"	Informational — those doctors are prioritised for a Friday night next month.

Good habits

- Keep the doctor list current — toggle people **Off** rather than deleting them for short absences.
- Clear stale **specific-date** demands each cycle so they don't fire in the wrong month.
- Generate, read the Summary, then regenerate if the balance looks off — only **Save** when you're happy.
- Download a **backup** before big edits.
- Rotate roster-maker passwords every cycle and set expiry dates.

Common questions

Question	Answer
A password won't work.	It may have expired, or been deactivated. Ask Dr. Alif to issue a new one.
I can't see the secret team.	Only the Master can. Roster makers never see secret doctors.
My saved roster disappeared after regenerating.	Regenerating replaces the in-memory roster. Re-save, or generate first and save the version you want.
Fridays aren't in my holiday list.	That's expected — Fridays are handled automatically and never need adding.

SECTION 16

Glossary

Term	Definition
SMO / EMO / MO	Senior / Emergency / Medical Officer — doctor categories that drive eligibility and targets.

Term	Definition
First Man	A senior on-call category restricted to priority wards.
Station	A staffing requirement: a group of wards needing N doctors on a given shift.
Demand	A standing request (off, double, single, leave, or fixed assignment) the generator must respect.
Duty Bank	The month-over-month ledger that carries over/under-worked duties into next month's targets.
Effective Target	A doctor's monthly target after Duty-Bank adjustment.
Secret team	Doctors visible only to the Master.
Master	The full-control role (Dr. Alif).
Roster Maker	A time-limited role for preparing rosters.
PWA	Progressive Web App — installable, works offline.

NHF&RI Doctor's Duty Roster App — User Manual. For access issues or a new roster-maker password, contact Dr. Alif.